

Here's a **proper exam-oriented 10-mark answer** you can write 📌

✍️ Features of Macros with Suitable Examples

◆ Introduction

- 👉 A **macro** is a named sequence of instructions used to perform a specific task.
 - 👉 It is defined once and can be used multiple times in a program.
 - 👉 During **macro expansion**, the macro call is replaced by its definition.
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◆ Features of Macros

1. ◆ Macro Substitution (Expansion)

- 👉 Macro name is replaced by its body.

Example:

```
MACRO  
CONST  
MOVER AREG, =5  
MEND
```

Call:

```
CONST
```

Expansion:

```
MOVER AREG, =5
```

2. ◆ Parameterized Macros

👉 Macros can accept parameters.

Example:

```
MACRO  
ADDNUM &A, &B  
MOVER AREG, &A  
ADD AREG, &B  
MOVEM AREG, &A  
MEND
```

Call:

```
ADDNUM X, Y
```

Expansion:

```
MOVER AREG, X  
ADD AREG, Y  
MOVEM AREG, X
```

3. ♦ Nested Macros (Macro Calls within Macro)

👉 A macro can call another macro.

Example:

```
MACRO  
INCR &X  
ADD &X, =1  
MEND
```

```
MACRO  
DOUBLE &X  
INCR &X  
INCR &X  
MEND
```

Call:

```
DOUBLE A
```

Expansion:

```
ADD A, =1  
ADD A, =1
```

4. ♦ Conditional Macro Expansion

👉 Execution depends on a condition.

Example:

```
MACRO  
CHECK &A  
IF &A GT 0  
MOVER AREG, &A  
ENDIF  
MEND
```

5. ♦ Looping in Macros

👉 Macros can repeat instructions using loops.

Example:

```
MACRO  
REPEAT &N  
LCL &I  
&I SET 1  
WHILE &I LE &N  
MOVER AREG, &I  
&I SET &I + 1  
ENDW  
MEND
```

6. ♦ Use of Local Variables

👉 Avoid conflicts using local variables.

Example:

```
MACRO  
TEMPADD &A, &B  
LCL &T  
&T SET &A
```

ADD &T, &B
MEND

◆ Advantages

- Reduces code duplication
 - Improves readability
 - Faster execution
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◆ Disadvantages

- Difficult debugging
 - No type checking
 - Code size may increase
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◆ Conclusion

👉 Macros provide powerful features like parameterization, nesting, and conditional execution, making programs more efficient and modular.

🎯 One-line Answer (Exam)

👉 *Macros support features like substitution, parameter passing, nesting, conditional and iterative expansion, making programs reusable and efficient.*

If you want, I can give you **macro processor tables (MNT, MDT, ALA)** which are very **important for exams** 👍